

Two Way ANOVA : More than one observation per cell.

Group of people	Drug		
	x	y	z
A	14	10	11
	15	9	11
B	12	7	10
	11	8	11
C	10	11	8
	11	11	7

Do the drugs act differently? Are the different group of people affected differently?
Is the interaction term significant?

$$i) T = 187 ; n = 18 ; C_T = \frac{187 \times 187}{18} = \underline{\underline{1942.72}}$$

$$ii) SST = \left[(14)^2 + (15)^2 + (12)^2 + (11)^2 + (10)^2 + (11)^2 + (10)^2 + (9)^2 + (7)^2 + (8)^2 + (11)^2 + (11)^2 + (11)^2 + (11)^2 + (10)^2 + (11)^2 + (8)^2 + (7)^2 \right] - C_T$$

$$= 2019 - 1942.72 = \underline{\underline{76.28}}$$

$$\begin{aligned}
 \text{iii) } SS B_c &= \left[\frac{73 \times 73}{6} + \frac{56 \times 56}{6} + \frac{58 \times 58}{6} \right] - C_F \\
 &= 888.16 + 522.66 + 560.67 - 1542.72 \\
 &= \underline{\underline{28.77}}
 \end{aligned}$$

$$\begin{aligned}
 \text{iv) } SS B_R &= \left[\frac{70 \times 70}{6} + \frac{59 \times 59}{6} + \frac{58 \times 58}{6} \right] - C_F \\
 &= (816.67 + 580.16 + 560.67) - 1542.72 \\
 &= \underline{\underline{14.78}}
 \end{aligned}$$

$$\begin{aligned}
 \text{v) } SS W_s &= (14 - 14.5)^2 + (15 - 14.5)^2 + (10 - 9.5)^2 + (9 - 9.5)^2 \\
 &\quad + (11 - 11)^2 + (11 - 11)^2 + (12 - 11.5)^2 + (11 - 11.5)^2 + (7 - 7.5)^2 \\
 &\quad + (8 - 7.5)^2 + (10 - 10.5)^2 + (11 - 10.5)^2 + (10 - 10.5)^2 \\
 &\quad + (11 - 10.5)^2 + (11 - 11)^2 + (11 - 11)^2 + (8 - 7.5)^2 + (7 - 7.5)^2 \\
 &= \underline{\underline{3.50}}
 \end{aligned}$$

$$\text{vi) } SST = SST - \{SS B_c + SS B_R + SS W_s\}$$

$$= 76.28 - [28.77 + 14.78 + 3.50]$$

$$= \underline{\underline{29.23}}$$

Source	SS	DF	MS	F-Ratio	F _α
B/W _c	28.77	(3-1)=2	28.77/2 = 14.385	$\frac{MSB_c}{MSW_s} = 36.5$	F(2,9) = 4.25
B/W _R	14.78	(3-1)=2	14.78/2 = 7.390	$\frac{MSB_R}{MSW_c} = 15.0$	F(2,9) = 4.26
W _s	3.50	17-2-2-4 = 9	$\frac{3.50}{9} = 0.389$		
Tot	29.23	4	$\frac{29.23}{4} = 7.308$	$\frac{MSE}{MSW_s} = 18.79$	F(4,9) = 3.63
Total	76.28	(18-1)=17			

DF2	DF1	α = 0.05	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	Inf
1	161.45		199.5	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.1	251.14	252.2	253.25	254.31	
2	18.513		19	19.164	19.247	19.296	19.33	19.353	19.371	19.385	19.396	19.413	19.429	19.446	19.454	19.462	19.471	19.479	19.487	19.496	
3	10.128		9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123	8.7855	8.7446	8.7029	8.6602	8.6385	8.6166	8.5944	8.572	8.5494	8.5264	
4	7.7086		6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.041	5.9988	5.9644	5.9117	5.8578	5.8025	5.7744	5.7459	5.717	5.6877	5.6581	5.6281	
5	6.6079		5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725	4.7351	4.6777	4.6188	4.5581	4.5272	4.4957	4.4638	4.4314	4.3985	4.365	
6	5.9874		5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.099	4.06	3.9999	3.9381	3.8742	3.8415	3.8082	3.7743	3.7398	3.7047	3.6689	
7	5.5914		4.7374	4.3468	4.1203	3.9715	3.866	3.787	3.7257	3.6767	3.6365	3.5747	3.5107	3.4445	3.4105	3.3758	3.3404	3.3043	3.2674	3.2298	
8	5.3177		4.459	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881	3.3472	3.2839	3.2184	3.1503	3.1152	3.0794	3.0428	3.0053	2.9669	2.9276	
→ 9	5.1174		4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789	3.1373	3.0729	3.0061	2.9365	2.9005	2.8637	2.8259	2.7872	2.7475	2.7067	
10	4.9646		4.1028	3.7083	3.478	3.3258	3.2172	3.1355	3.0717	3.0204	2.9782	2.913	2.845	2.774	2.7372	2.6996	2.6609	2.6211	2.5801	2.5379	
11	4.8443		3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.948	2.8962	2.8536	2.7876	2.7186	2.6464	2.609	2.5705	2.5309	2.4901	2.448	2.4045	
12	4.7472		3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964	2.7534	2.6866	2.6169	2.5436	2.5055	2.4663	2.4259	2.3842	2.341	2.2962	
13	4.6672		3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144	2.671	2.6037	2.5331	2.4589	2.4202	2.3803	2.3392	2.2966	2.2524	2.2064	
14	4.6001		3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458	2.6022	2.5342	2.463	2.3879	2.3487	2.3082	2.2664	2.2229	2.1778	2.1307	
15	4.5431		3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876	2.5437	2.4753	2.4034	2.3275	2.2878	2.2468	2.2043	2.1601	2.1141	2.0658	
16	4.494		3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377	2.4935	2.4247	2.3522	2.2756	2.2354	2.1938	2.1507	2.1058	2.0589	2.0096	
17	4.4513		3.5915	3.1968	2.9647	2.81	2.6987	2.6143	2.548	2.4943	2.4499	2.3807	2.3077	2.2304	2.1898	2.1477	2.104	2.0584	2.0107	1.9604	
18	4.4139		3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117	2.3421	2.2686	2.1906	2.1497	2.1071	2.0629	2.0166	1.9681	1.9168	
19	4.3807		3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779	2.308	2.2341	2.1555	2.1141	2.0712	2.0264	1.9795	1.9302	1.878	
20	4.3512		3.4928	3.0984	2.8661	2.7109	2.599	2.514	2.4471	2.3928	2.3479	2.2776	2.2033	2.1242	2.0825	2.0391	1.9938	1.9464	1.8963	1.8432	
21	4.3248		3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.366	2.321	2.2504	2.1757	2.096	2.054	2.0102	1.9645	1.9165	1.8657	1.8117	
22	4.3009		3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419	2.2967	2.2258	2.1508	2.0707	2.0283	1.9842	1.938	1.8894	1.838	1.7831	
23	4.2793		3.4221	3.028	2.7955	2.64	2.5277	2.4422	2.3748	2.3201	2.2747	2.2036	2.1282	2.0476	2.005	1.9605	1.9139	1.8648	1.8128	1.757	
24	4.2597		3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002	2.2547	2.1834	2.1077	2.0267	1.9838	1.939	1.892	1.8424	1.7896	1.733	
25	4.2417		3.3852	2.9912	2.7587	2.603	2.4904	2.4047	2.3371	2.2821	2.2365	2.1649	2.0889	2.0075	1.9643	1.9192	1.8718	1.8217	1.7684	1.711	
26	4.2252		3.369	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655	2.2197	2.1479	2.0716	1.9898	1.9464	1.901	1.8533	1.8027	1.7488	1.6906	
27	4.21		3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501	2.2043	2.1323	2.0558	1.9736	1.9299	1.8842	1.8361	1.7851	1.7306	1.6717	
28	4.196		3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.236	2.19	2.1179	2.0411	1.9586	1.9147	1.8687	1.8203	1.7689	1.7138	1.6541	
29	4.183		3.3277	2.934	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229	2.1768	2.1045	2.0275	1.9446	1.9005	1.8543	1.8055	1.7537	1.6981	1.6376	
30	4.1709		3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646	2.0921	2.0148	1.9317	1.8874	1.8409	1.7918	1.7396	1.6835	1.6223	
40	4.0847		3.2317	2.8387	2.606	2.4495	2.3359	2.249	2.1802	2.124	2.0772	2.0035	1.9245	1.8389	1.7929	1.7444	1.6928	1.6373	1.5766	1.5089	

28	4.196	3.3404	2.9467	2.7141	2.5581	2.4433	2.3593	2.2913	2.236	2.19	2.1179	2.0411	1.9586	1.9147	1.8667	1.8203	1.7689	1.7138	1.6541
29	4.183	3.3277	2.934	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229	2.1768	2.1045	2.0275	1.9446	1.9005	1.8543	1.8055	1.7537	1.6981	1.6376
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646	2.0921	2.0148	1.9317	1.8874	1.8409	1.7918	1.7396	1.6835	1.6223
40	4.0847	3.2317	2.8387	2.606	2.4495	2.3359	2.249	2.1802	2.124	2.0772	2.0035	1.9245	1.8389	1.7929	1.7444	1.6928	1.6373	1.5766	1.5089
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.097	2.0401	1.9926	1.9174	1.8364	1.748	1.7001	1.6491	1.5943	1.5343	1.4673	1.3893
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.175	2.0868	2.0164	1.9588	1.9105	1.8337	1.7505	1.6587	1.6084	1.5543	1.4952	1.429	1.3519	1.2539
Inf	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799	1.8307	1.7522	1.6664	1.5705	1.5173	1.4591	1.394	1.318	1.2214	1

Verdict: Reject all the null hypothesis.

= Kruskal-Wallis test: Analogous to 1-way ANOVA.

H_0 : 'k' independent random samples come from identical universes.

H_1 : Medians of these universes are not equal.

$$\Rightarrow H = \frac{12}{n(n+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(n+1)$$

where $n = n_1 + n_2 + \dots + n_k$ & R_i being the sum of ranks assigned to n_i observations in the i^{th} sample.

$\Rightarrow$ If H_0 is true, there is no difference b/w sample medians & each sample has almost

sample medians & each sample was repeated
5 times, then sampling distribution of $\bar{X}$
can be approximated with a Chi-Square
distribution with $(k-1)$ dof.